

AI-ML Assignment: Exploratory Data Analysis (EDA) Project

Project Overview

The goal of this project is to apply exploratory data analysis (EDA) techniques to a real-world dataset of your choice. You will take on the role of a Data Analyst to uncover hidden patterns, clean messy data, evaluate variable relationships, and present data-driven insights.

Deliverables

You must submit exactly two files:

1. [YourName]_Project.ipynb: A clean, fully executed, and well-commented Jupyter Notebook.
2. [YourName]_Report.pdf: A formal technical report generated from your notebook or compiled separately, containing your visualizations, findings, and business conclusions.

Part 1: Project Checklist & Workflow

Your project must strictly follow these five phases of analysis:

Phase 1: Dataset Selection & Problem Definition

- Go to **Kaggle** (or any open-data repository) and download a dataset of interest.
- *Constraint:* The dataset must contain at least **500 rows** and a mix of both **numerical** and **categorical** variables.
- In your introduction, clearly state:
 - The source of the data.
 - What each column represents (Data Dictionary).
 - 3 to 5 core business questions you hope to answer using this data.

Phase 2: Data Cleaning & Integrity Checks

- Check your data types (.info()) and fix columns loaded as the wrong format (e.g., dates loaded as text strings).
- Identify and handle **Missing Values**: Decide whether to drop them or impute them (using mean/median/mode) and justify your choice.
- Check for and remove **Duplicate Rows**.

Phase 3: Outlier Analysis

- Isolate numerical columns and check their distributions using **Skewness** metrics and **Box Plots**.
- Write an IQR (Interquartile Range) loop to detect outliers across your continuous variables.



- Apply an appropriate handling strategy: **Trimming** (dropping), **Capping** (Winsorization), or **Transformation** (Log/Sqrt scaling) based on what is healthiest for your data distribution.

Phase 4: Bivariate & Multivariate Exploration

- Construct a **Correlation Heatmap** of all numerical features. Explain the strong positive and negative relationships you observe.
- Use **Box Plots** or **Bar Charts** to cross-examine categorical variables against numeric ones (e.g., comparing *Income* across different *Job Types*).
- State clearly whether your observed correlations imply direct causation or if a hidden variable might be at play.

Phase 5: Feature Engineering & Encoding Preparation

- Create at least one new feature column that combines existing data logically (e.g., calculating *Car_Age* from a manufacturing year).
- Identify which categorical columns are **Ordinal** (ranked) and which are **Nominal** (unranked).
- Demonstrate preparation for predictive modeling by applying **Label Encoding** to ranked categories and **One-Hot Encoding** (Dummy variables) to unranked ones.

Part 2: Submission Instructions

- **Late Submission Policy:** Submissions close strictly on the deadline. Late submissions lose 10% of total marks per day.
- **Format:** Zip your .ipynb and .pdf files together into a single file named [YourName]_EDA_Project.zip.
- **Academic Integrity:** All code and text must be your own work. Group work or uncredited code copying will result in an immediate failure of the assignment.

